

HOMI BHABHA BALVAIDNYANIK COMPETITION

Class: 6 • Subject: Science • Topic: Work and Energy

PART A – WORK

1. What is Work?

In science, work is said to be done when a force acting on an object causes the object to move through a distance in the direction of the force.

Formula

$$W = F \times d$$

Where:

- W = Work done
- F = Force applied
- d = Distance moved in the direction of the force

SI Unit

The SI unit of work is the joule (J).

$$1 \text{ J} = 1 \text{ N} \times 1 \text{ m}$$

Therefore:

1 joule is the work done when a force of 1 newton moves an object through 1 metre in the direction of the force.

2. When is Work Done?

There are two important conditions:

1. A force must act on the object.
2. The object must undergo displacement in the direction of the force.

Example

A force of 20 N moves a box through 5 m.

$$W = F \times d$$

$$W = 20 \times 5 = 100\text{J}$$

Therefore, 100 J of work is done.

3. When is No Work Done?

If an object does not move despite a force being applied, the work done on the object is zero.

Example

A person pushes a wall with great force, but the wall does not move.

$$d=0$$

Therefore:

$$W=F \times 0=0$$

Work done = 0 J

Important Examination Concept

Force alone does not necessarily mean work is done.

There must be movement/displacement.

4. Work Done in Different Situations

Situations	Is work done?
Pushing a box and it moves	Yes
Lifting a book upward	Yes
Pushing a wall that does not move	No
Holding a heavy bag stationary	No mechanical work on the bag due to the holding force
Moving a trolley	Yes
Kicking a football and making it move	Yes

PART B – ENERGY

5. What is Energy?

Energy is the capacity or ability to do work.

The SI unit of energy is also the joule (J).

Important Relationship

Energy allows work to be done.

For example:

- A moving ball has energy and can do work when it hits another object.
- A raised stone has stored energy and can do work when it falls.

PART C – FORMS OF ENERGY

Some important forms of energy are:

1. Kinetic energy
2. Potential energy
3. Heat energy
4. Light energy
5. Sound energy
6. Electrical energy
7. Chemical energy
8. Wind energy

For this topic, the most important forms are kinetic energy and potential energy.

PART D – KINETIC ENERGY

6. What is Kinetic Energy?

Kinetic energy is the energy possessed by an object due to its motion.

Examples

- A moving car
- A running athlete
- A flying bird
- A rolling ball
- Flowing water
- Moving air

Key Idea

Moving object → Kinetic Energy

A stationary object has zero kinetic energy in the usual school-level treatment.

7. Factors Affecting Kinetic Energy

Kinetic energy depends mainly on:

① Mass

For objects moving at the same speed:

Greater mass → Greater kinetic energy

② Velocity

For objects having the same mass:

Greater velocity → Greater kinetic energy

The mathematical relationship is:

$$KE = \frac{1}{2}mv^2$$

where:

- KE = kinetic energy
- m = mass
- v = velocity

8. Important Effect of Velocity

Notice that velocity is squared.

$$KE \text{ proportional to } v^2$$

Therefore:

- If velocity doubles → KE becomes 4 times
- If velocity triples → KE becomes 9 times
- If velocity becomes half → KE becomes 1/4

This is a very important numerical and conceptual point.

PART E – COMPARING KINETIC ENERGIES

9. Comparison Without Full Calculation

Example 1

Object A:

- Mass = 2 kg
- Velocity = 5 m/s

Object B:

- Mass = 4 kg
- Velocity = 5 m/s

Both have the same velocity.

Object B has twice the mass.

Therefore:

$$\text{KE of B} = 2 \times \text{KE of A}$$

Example 2

Object A:

- Mass = 2 kg
- Velocity = 4 m/s

Object B:

- Mass = 2 kg
- Velocity = 8 m/s

Mass is the same, but velocity of B is double.

Therefore:

$$\text{KE}_B = 2^2 \times \text{KE}_A$$

$$\text{KE}_B = 4\text{KE}_A$$

So B has four times the kinetic energy.

PART F – POTENTIAL ENERGY

10. What is Potential Energy?

Potential energy is the energy possessed by an object due to its position, condition or configuration.

It is often described as stored energy.

Examples

- A stone held at a height
- Water stored in a dam
- A stretched rubber band
- A compressed spring
- A drawn bow

PART G – GRAVITATIONAL POTENTIAL ENERGY

11. Potential Energy Due to Height

When an object is raised above the ground, it gains gravitational potential energy.

For school-level numericals:

$$\text{PE} = mgh$$

where:

- PE = potential energy
- m = mass
- g = acceleration due to gravity
- h = height

For many school problems, use:

$$g = 10 \text{ m/s}^2$$

unless another value is given.

12. Factors Affecting Potential Energy

Gravitational potential energy depends on:

① Mass

For the same height:

Greater mass → Greater PE

② Height

For the same mass:

Greater height → Greater PE

③ Gravitational acceleration

PE proportional to g

PART H – COMPARING POTENTIAL ENERGIES

13. Comparison Based on Mass

Object A:

- Mass = 2 kg
- Height = 5 m

Object B:

- Mass = 4 kg
- Height = 5 m

Height is the same.

Mass of B is twice that of A.

Therefore:

$$PE_B = 2PE_A$$

14. Comparison Based on Height

Object A:

- Mass = 5 kg
- Height = 2 m

Object B:

- Mass = 5 kg
- Height = 6 m

Mass is the same.

Height of B is three times that of A.

Therefore:

$$PE_B = 3PE_A$$

15. Comparison When Both Mass and Height Change

Object A:

- Mass = 2 kg
- Height = 4 m

Object B:

- Mass = 4 kg
- Height = 8 m

Mass doubles and height doubles.

Therefore:

$$PE_B = 2 \times 2 \times PE_A$$

$$PE_B = 4PE_A$$

PART I – KINETIC AND POTENTIAL ENERGY

16. Difference Between Kinetic and Potential Energy

Kinetic Energy	Potential Energy
Energy due to motion	Energy due to position/condition
Requires movement	Can exist without movement
Depends on mass and velocity	Gravitational PE depends on mass, height and g
Example: moving car	Example: raised stone
Formula: $\frac{1}{2}mv^2$	Formula: mgh

Easy Concept

Kinetic = Motion

Potential = Position/Stored

PART J – ENERGY CONVERSION

17. What is Energy Conversion?

Energy conversion is the transformation of energy from one form to another.

Energy does not simply disappear; it can change from one form to another.

18. Important Examples

Falling Object

At the top:

High Potential Energy → Low Kinetic Energy

As it falls:

Potential Energy → Kinetic Energy

Just before reaching the ground, its kinetic energy is greatest if air resistance is ignored.

Electric Bulb

Electrical Energy → Light Energy + Heat Energy

Electric Fan

Electrical Energy $\rightarrow$ Mechanical/Kinetic Energy + Sound + Heat

Battery-Powered Toy Car

Chemical Energy $\rightarrow$ Electrical Energy $\rightarrow$ Mechanical/Kinetic Energy + Sound + Heat

Food in Our Body

Chemical Energy $\rightarrow$ Kinetic Energy + Heat Energy

Hydroelectric Power Plant

Potential Energy of Stored Water $\rightarrow$ Kinetic Energy of Moving Water $\rightarrow$ Mechanical Energy $\rightarrow$ Electrical Energy

Wind Turbine

Kinetic Energy of Wind $\rightarrow$ Mechanical Energy $\rightarrow$ Electrical Energy

PART K – CONSERVATION OF ENERGY

19. Law of Conservation of Energy

Energy can neither be created nor destroyed; it can only be transformed from one form to another.

The total amount of energy remains conserved, although its form may change.

Example

When a ball falls:

Potential Energy $\rightarrow$ Kinetic Energy

The energy changes form.

PART L – WORK AND ENERGY CONNECTION

20. Relationship Between Work and Energy

When work is done on an object, energy can be transferred to the object.

For example:

A person pushes a stationary box.

The applied force moves the box.

Work is done on the box, transferring energy to it.

Similarly, when an object does work, it transfers some of its energy to another object.

Important Relationship

Work is a way of transferring energy.

PART M – NUMERICAL PRACTICE

A. Work Done

1.

A force of 25 N moves an object through 8 m.

Find the work done.

2.

A force of 60 N moves a box through 5 m.

Calculate the work done.

3.

A force does 500 J of work while moving an object through 10 m.

Find the force.

4.

A force of 40 N does 240 J of work.

Find the distance moved.

5. Conceptual Numerical

A student pushes a wall with a force of 150 N, but the wall does not move.

Calculate the work done.

B. Kinetic Energy

Use:

$$KE = \frac{1}{2}mv^2$$

6.

Calculate the kinetic energy of an object of mass 4 kg moving at 5 m/s.

7.

An object of mass 2 kg moves at 10 m/s.

Find its kinetic energy.

8. Comparison

Object A has a mass of 2 kg and moves at 4 m/s.

Object B has a mass of 4 kg and moves at 4 m/s.

Which object has greater kinetic energy and how many times greater is it?

9. Comparison

Two objects have equal masses.

Object A moves at 5 m/s, while object B moves at 10 m/s.

Compare their kinetic energies.

10. Challenge

Object A has mass 4 kg and velocity 3 m/s.

Object B has mass 2 kg and velocity 6 m/s.

Compare their kinetic energies.

C. Potential Energy

Use:

$$PE = mgh$$

Take:

$$g = 10 \text{ m/s}^2$$

11.

A 5 kg object is kept at a height of 4 m.

Calculate its gravitational potential energy.

12.

A 2 kg object is raised to a height of 10 m.

Find its potential energy.

13. Comparison

Object A has a mass of 2 kg and is at a height of 5 m.

Object B has a mass of 4 kg and is at the same height.

Compare their potential energies.

14. Comparison

Two objects have equal masses.

Object A is at a height of 2 m, while object B is at 6 m.

Compare their potential energies.

15. Challenge

Object A has mass 2 kg and height 4 m.

Object B has mass 4 kg and height 8 m.

How many times is the potential energy of B greater than that of A?

D. Energy Conversion**16.**

A ball is held at a height and then released.

What energy conversion mainly occurs as it falls?

17.

Identify the main energy conversion in an electric fan.

18.

Identify the energy conversion in a hydroelectric power station, starting with water stored at a height.

19.

A battery-operated toy car starts moving after its switch is turned on.

Write the main sequence of energy conversions.

20. Higher-Level Concept

A ball is thrown vertically upward.

(a)

What happens to its kinetic energy as it moves upward?

(b)

What happens to its potential energy?

(c)

At the highest point, which type of mechanical energy is greatest?

ANSWERS & SOLUTIONS

Answer Key Preview

1. 200 J
2. 300 J
3. 50 N
4. 6 m
5. 0 J
6. 50 J
7. 100 J
8. B; 2 times A
9. B; 4 times A
10. Equal kinetic energies
11. 200 J
12. 200 J
13. B; 2 times A
14. B; 3 times A
15. 4 times
16. Potential → Kinetic
17. Electrical → Mechanical/Kinetic + Sound + Heat
18. Potential → Kinetic → Mechanical → Electrical
19. Chemical → Electrical → Mechanical/Kinetic + Sound + Heat
20. (a) Kinetic energy decreases (b) Potential energy increases (c) Potential energy is greatest